National University of Singapore

DBA5101

Group Project 2

Recycling Behaviour at NUS

Bin Openings and Informational Banners

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Executive Summary

This study evaluated whether reshaped recycling-bin openings and informational banners reduced contamination at UTOWN, compared with unchanged bins at ENGINE. The analysis examined paper, plastic, and can recyclables across three trial phases.

The reshaped openings worked best on can bins. The share of wrong items in UTOWN's can bins fell from 35% to 18% while the openings were in place, about 15 percentage points more than the fall at ENGINE, which suggests a significant effect. However, the openings had no clear effect on plastic or paper. Plastic contamination fell by about 10 points at both UTOWN and ENGINE, so that fall cannot be credited to solely the openings. Paper bins were already fairly clean and changed little. The banners did not clearly reduce contamination for any material, and plastic contamination at UTOWN went back up to 59%, slightly above where it started.

The team recommends keeping the reshaped openings on can bins and testing them at more locations, dropping the banners, and running a longer test on plastic, the most contaminated bin type. The can result is promising rather than proven. The gap between the two areas' can bins was already narrowing before the openings were installed, and measuring from 27 January rather than mid-January reduces the estimated effect from 15 to about 10 points. The two test periods each lasted two weeks or less, the trial compares a single residential area with a single academic area, and the data do not record how much was recycled or what sorting cost.

1 The Trial and the Question

Recycling bins at NUS often contain items that do not belong in them, such as food packaging in a paper bin. This contamination makes sorting more costly and can cause a whole load to be sent to general waste. In early 2020, NUS tested two low-cost ways of reducing it at University Town (UTOWN), a residential area of campus. For the first four weeks the bins were left unchanged, to record normal behavior. For the next twelve days, the bin openings were reshaped to match the item each bin is meant for. By March, the reshaped openings had been removed, and banners showing common recycling mistakes were hung above the bins for two weeks. Bins in the engineering area (ENGINE) were left unchanged throughout. They show how contamination would have moved without any changes, for example as the semester progressed, so each change at UTOWN is measured against the change at ENGINE over the same days.

For each material, the team asks whether contamination at UTOWN fell by more than at ENGINE over the same days. This comparison is called a difference-in-differences (DiD). It takes the change at UTOWN and subtracts the change at ENGINE, so that anything that affects both areas in the same way, such as the time of the semester, cancels out. The question is asked for each of the two test periods, the period with the reshaped openings (Phase 2) [9] and the period with the banners (Phase 3) [10]. Both are compared with the first four weeks (Phase 1), when neither area had any changes. Phase 3 is not mainly compared with Phase 2, because the openings were removed when the banners went up, so that comparison would mix the two changes and the 12-day openings period was too short to form lasting habits, so the Phase 3 estimate is read as the effect of the banners alone. Plastic returning to its baseline level supports this; for cans, which stayed below baseline, any persisting habit would make the banners look more effective than they were. Table 1 sets out the three phases.

Phase	Dates, 2020	UTOWN	ENGINE
1	13 January to 10 February	Standard bins	Standard bins
2	11 to 22 February	Reshaped openings	Standard bins
3	4 to 17 March	Banners, openings removed	Standard bins

Table 1: The three phases of the trial.

UTOWN contains housing, sports facilities and dining, while ENGINE contains classrooms, social areas and dining. The published case study of the trial describes nine bin sites in UTOWN and seven in ENGINE, but the data give one daily figure for each area. The trial therefore compares one changed area with one unchanged area, rather than individual bin sites. [1, 3, 2]

2 The Data

The data file records, for each area and day, the percentage of wrong items found in its paper, plastic and can bins. After 170 empty rows are removed, it contains 85 daily records, 43 from ENGINE and 42 from UTOWN, between 13 January and 17 March 2020. No day appears twice for the same area. Recorded zeros are kept as they are, and missing readings are not filled in.

The team uses only days on which both areas were recorded, so that UTOWN and ENGINE are always compared on the same day. This leaves 18 days in Phase 1 (17 for cans), 11 in Phase 2 and 11 in Phase 3. Each day counts equally, because the file does not record how many items were collected on each day, so the averages describe a typical day rather than the share of all items collected.

Two sets of unusual readings, on 2 February and on 12 and 16 March, are kept in the main results and examined in Section 6.

3 Measuring the Effect

As set out in Section 1, the DiD subtracts the change at ENGINE from the change at UTOWN. For each material and each test period, it is calculated from four averages, UTOWN before and after and ENGINE before and after. The result is in percentage points (pp), the difference between two percentages, so a value of −15 means contamination at UTOWN fell 15 points more than at ENGINE. A negative value is an improvement at UTOWN.

The same number is obtained from the regression below, which also gives a 95% interval and a p-value.

$$Y = \alpha + \beta\, UTOWN + \gamma\, After + \delta\,(UTOWN \times After) + \varepsilon$$

Here Y is one day's contamination at one area. *UTOWN* equals 1 for UTOWN and 0 for ENGINE. *After* equals 1 in the test period and 0 in Phase 1. The coefficient δ is the DiD. If its 95% interval excludes zero, the change is unlikely to be due to chance. An interval that includes zero means the data cannot rule out no change, not that the change had no effect. The interval assumes that each day's reading is independent of the next and that day-to-day variation is similar in both areas and periods. Neither may hold for repeated readings from the same bins [11], as the spread checks in Appendix C show. The interval also does not allow for the trend concern in Section 4. Appendix A gives the four averages for each comparison and the regression coefficients for cans.

4 Checking the Comparison Area

The comparison in Section 1 rests on one assumption. Without the changes, contamination at UTOWN and ENGINE would have moved in step, so the gap between them would have stayed roughly the same. This is known as the parallel trends assumption. It cannot be tested directly, because no one can see what UTOWN would have looked like without the changes. The team instead checks whether the gap was already moving during Phase 1, before anything was changed. Figure 1 plots the daily gap, UTOWN minus ENGINE, for each material over that period.

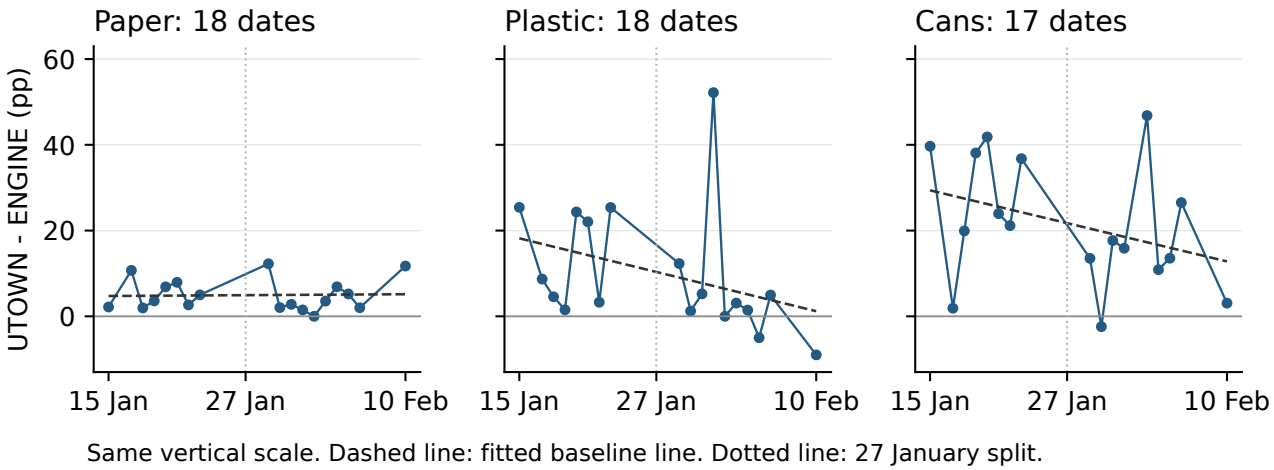


Figure 1: Daily UTOWN minus ENGINE gap during Phase 1.

For paper, the gap stayed flat. For plastic and cans, it was narrowing by about 4.5 percentage points a week. With so few days, these trends cannot be told apart from random day-to-day variation (Appendix A), but they matter for the can result. If the can gap had kept shrinking at that rate, it would account for most of the 15-point fall credited to the openings. Two findings weigh against reading the whole fall as a trend. The can gap widened again once the openings were removed (Figure 2), and measuring from 27 January, when the gap had already narrowed, still leaves an estimated effect of about 10 points (Section 6).

5 Results

Figure 2 and Table 2 use the matched days described in Section 2. In Figure 2, the dashed line shows where UTOWN would have been had it moved in step with ENGINE, and the arrow is the DiD. Table 2 reports these changes with their 95% intervals and p-values from the regression in Section 3. In Table 2, P2 and P3 denote the openings and banners periods.

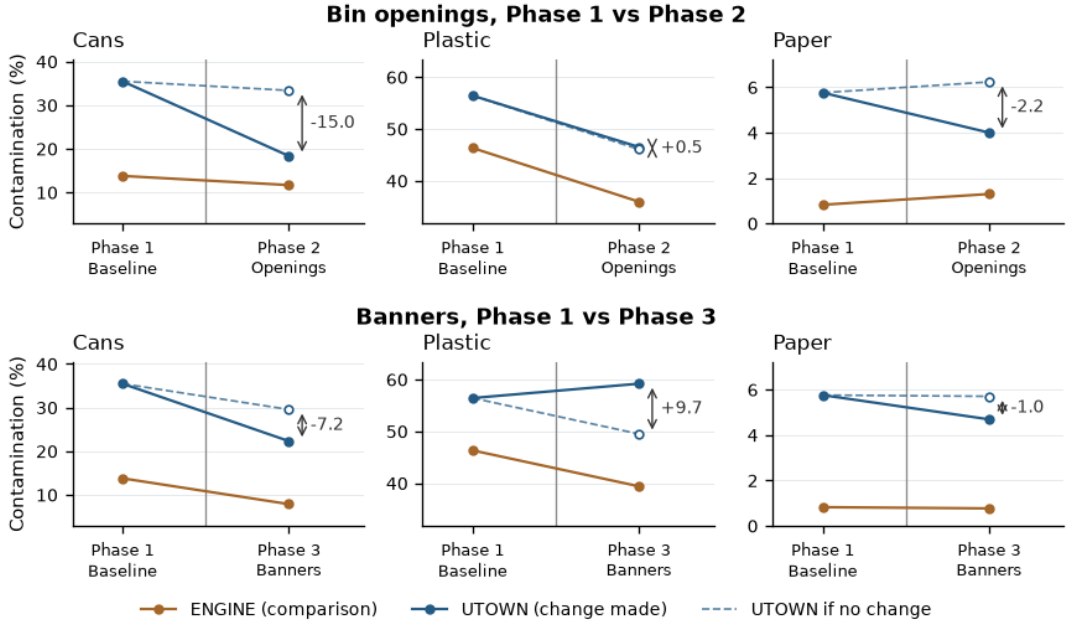


Figure 2: Average daily contamination in each area, with the estimated effect of each change.

Comparison	Material	Dates before/after	DiD (pp)	95% interval (pp)	p-value
P2 versus P1	Paper	18/11	-2.23	[-4.81, 0.34]	0.088
P2 versus P1	Plastic	18/11	+0.46	[-14.40, 15.31]	0.951
P2 versus P1	Cans	17/11	-15.02	[-25.73, -4.32]	0.007
P3 versus P1	Paper	18/11	-1.01	[-3.54, 1.51]	0.424
P3 versus P1	Plastic	18/11	+9.69	[-4.68, 24.07]	0.182
P3 versus P1	Cans	17/11	-7.24	[-16.78, 2.29]	0.133

Table 2: Main results. Each DiD compares the test period with Phase 1.

The can gap shows the clearest pattern. It fell from 21.7 points in Phase 1 to 6.7 under the openings, a DiD of −15.0 (interval −25.7 to −4.3, p = 0.007) and the only clear result in the trial. At UTOWN, can contamination fell from 35.5% to 18.3%, while at ENGINE it fell from 13.8% to 11.7%. Under the banners the gap rose back to 14.5 points, giving a DiD of −7.2 that is not clear (interval −16.8 to +2.3). The partial rebound once the openings were removed supports reading the Phase 2 fall as an effect of the openings. The trend concern in Section 4 still applies, and part of the Phase 3 level may reflect habits carried over from the openings period.

The plastic gap barely moved under the openings, from 10.1 to 10.6 points, a DiD of +0.5. Contamination fell by about 10 points at both UTOWN and ENGINE, so the fall cannot be credited to the openings, and the wide interval (−14.4 to +15.3) cannot rule out a moderate effect either way. Under the banners the gap widened to 19.8 points, as UTOWN rose from 56.4% to 59.2% while ENGINE fell from 46.3% to 39.4%. That DiD of +9.7 is not clear (interval −4.7 to +24.1), but plastic shows no sign of improvement under either change.

Paper contamination is low at both areas, and the gap stayed below 5 points throughout. The DiD is −2.2 under the openings (p = 0.088) and −1.0 under the banners (p = 0.424), so neither is clear. Most of the Phase 2 fall comes from four Phase 1 days on which UTOWN's paper contamination was above 8%, which did not occur in Phase 2 or Phase 3. Without those days the DiD is −0.6 points. Because such days also stayed away under the banners, their absence cannot be credited to the openings.

Taken together, the openings did at least as well as the banners on every material, with the largest differences for plastic and cans. Neither change improved plastic.

6 Robustness and Limits

The team repeats the DiD with different choices of days, to see whether the main results depend on them. Table 3 lists the results in percentage points, with the main analysis in the first row. In the table, U is UTOWN.

Two sets of readings look unusual. On 2 February, paper and plastic contamination are recorded as exactly zero at both areas and no can readings were recorded, which may mean no collection took place. UTOWN's readings on 12 and 16 March are identical in all three materials, which may be a copying error. Neither can be confirmed as a mistake, so the main results keep both, and two rows of Table 3 remove them. The file also includes weekend dates although the case study describes weekday collection, so one check uses weekdays only. 22 February is the first day of the mid-semester recess and the only recess day in the data, so another check drops it. The last check, the recent baseline, starts on 27 January, to reduce the effect of the narrowing gap described in Section 4.

Sample	P2 Paper	P2 Plastic	P2 Cans	P3 Paper	P3 Plastic	P3 Cans
Main matched	-2.23	+0.46	-15.02	-1.01	+9.69	-7.24
All readings	-2.52	-0.80	-14.96	-1.29	+6.96	-6.87
Exclude 2 Feb / U 16 Mar	-2.52	-0.14	-15.02	-1.58	+10.37	-7.79
Exclude 2 Feb / U 12 Mar	-2.52	-0.14	-15.02	-1.48	+9.71	-6.99
Weekdays only	-3.10	+0.91	-12.99	-1.69	+9.52	-8.28
Without 22 Feb	-2.17	+0.89	-12.28	-1.01	+9.69	-7.24
Recent baseline	-2.09	+3.90	-9.50	-0.87	+13.14	-1.72

Table 3: DiD estimates under alternative choices of days.

Every estimate keeps its direction across all versions, except plastic under the openings, which stays close to zero. The can estimate under the openings ranges from −12.3 to −15.0 points in every version except the recent baseline, where it falls to −9.5. With the recent baseline, the can estimate under the banners falls to −1.7, while plastic under the banners ranges from +7.0 to +13.1 across all versions. These are different comparisons, not upper and lower limits on the effect.

6.1 Outside Events and Missing Data

Two calendar events fall close to the phase changes. NUS moved classes of more than 50 students online from 10 February, one day before Phase 2 due to COVID-19. [7] The mid-semester recess ran from 22 February to 1 March, between Phases 2 and 3. [8] Either could have affected residential UTOWN and academic ENGINE differently, which the DiD cannot separate from the effect of the changes. Their effect on contamination was not measured, so the team cannot say how much of each estimate they explain. The data also do not record how much material was collected, how many people used the bins or what sorting cost, so a lower contamination rate does not show whether more was recycled or whether collection became cheaper.

7 Recommendation and Next Steps

The reshaped openings are the only change followed by a clear fall, about 15 points in can contamination at UTOWN relative to ENGINE. They showed no clear effect on plastic or paper, and the banners showed no clear improvement for any material. The team therefore recommends keeping the openings on can bins and extending them to more sites, dropping the banners, and testing plastic separately over a longer period.

Because the can result rests on a single area and a baseline that was already narrowing, the extension should be run as a better-designed trial. Comparable bin sites should be assigned at random [6] to openings or standard bins and observed over the same weeks, so that calendar events affect both groups alike. Each phase should run for at least 20 collection days, about twice the length of this trial, which would give a good chance of detecting a 10-point change in plastic, the most contaminated stream. A short period with no changes between interventions, a constant number of measured sites, and records of how much material is collected and what sorting costs would also show whether cleaner bins lead to more recycling at lower cost.

A Detailed Calculations

These tables show the arithmetic behind the main results. Tables 4 and 5 use the same matched days as Table 2 and give the four averages behind each DiD, in percent, with the DiD in percentage points. All calculations use unrounded values.

Material	UTOWN before	UTOWN after	ENGINE before	ENGINE after	DiD (pp)
Paper	5.76	4.00	0.82	1.29	-2.23
Plastic	56.41	46.51	46.32	35.96	0.46
Cans	35.47	18.34	13.77	11.67	-15.02

Table 4: Four averages behind each DiD, openings period versus Phase 1.

Material	UTOWN before	UTOWN after	ENGINE before	ENGINE after	DiD (pp)
Paper	5.76	4.69	0.82	0.77	-1.01
Plastic	56.41	59.18	46.32	39.39	9.69
Cans	35.47	22.31	13.77	7.86	-7.24

Table 5: Four averages behind each DiD, banners period versus Phase 1.

The baseline trend in Section 4 comes from a regression of the daily gap on the number of days since 13 January, multiplied by seven to give points per week. The placebo check applies the same DiD regression within Phase 1, treating 27 January as if it were a change date. Neither check can confirm parallel trends, because Phase 1 is short.

Material	Gap trend (pp/week)	Trend p	Placebo DiD (pp)	Placebo p
Paper	+0.11	0.892	-0.32	0.865
Plastic	-4.58	0.147	-7.76	0.486
Cans	-4.46	0.168	-11.74	0.079

Table 6: Baseline trend and placebo checks.

Table 7 shows the regression from Section 3 for cans in the openings period. The intercept is ENGINE's Phase 1 average, the UTOWN coefficient is the starting gap between the areas, the After coefficient is ENGINE's change, and the interaction is UTOWN's extra change, which is the DiD.

Term	Coefficient	Standard error	p-value
Intercept	13.77	2.37	0.000
UTOWN	21.69	3.34	0.000
After	-2.10	3.77	0.581
UTOWN x After	-15.02	5.34	0.007

Note. A displayed p-value of 0.000 is rounded, not literally zero.

Table 7: Regression coefficients for cans, openings period versus Phase 1.

Material	UTOWN change (pp)	ENGINE change (pp)	Relative change (pp)
Paper	+0.69	-0.53	+1.22
Plastic	+12.67	+3.44	+9.24
Cans	+3.96	-3.82	+7.78

Table 8: Direct comparison of the banners period with the openings period (Section 6). This is not a separate banner effect.

B Data Checks and Daily Readings

Table 9 gives each area's average in each phase using every recorded day, and Figure 3 plots the daily readings used in the main results.

Phase	Area	Dates	Paper (%)	Plastic (%)	Cans (%)	Can dates
1	ENGINE	20	0.81	45.94	14.85	19
1	UTOWN	20	6.04	57.28	36.47	19
2	ENGINE	11	1.29	35.96	11.67	11
2	UTOWN	11	4.00	46.51	18.34	11
3	ENGINE	12	0.75	40.88	7.55	12
3	UTOWN	11	4.69	59.18	22.31	11

Table 9: Averages using all recorded days, matching Exhibit 5 of the case study. These differ slightly from the matched-day averages in the main results. Can dates count days with a can reading.

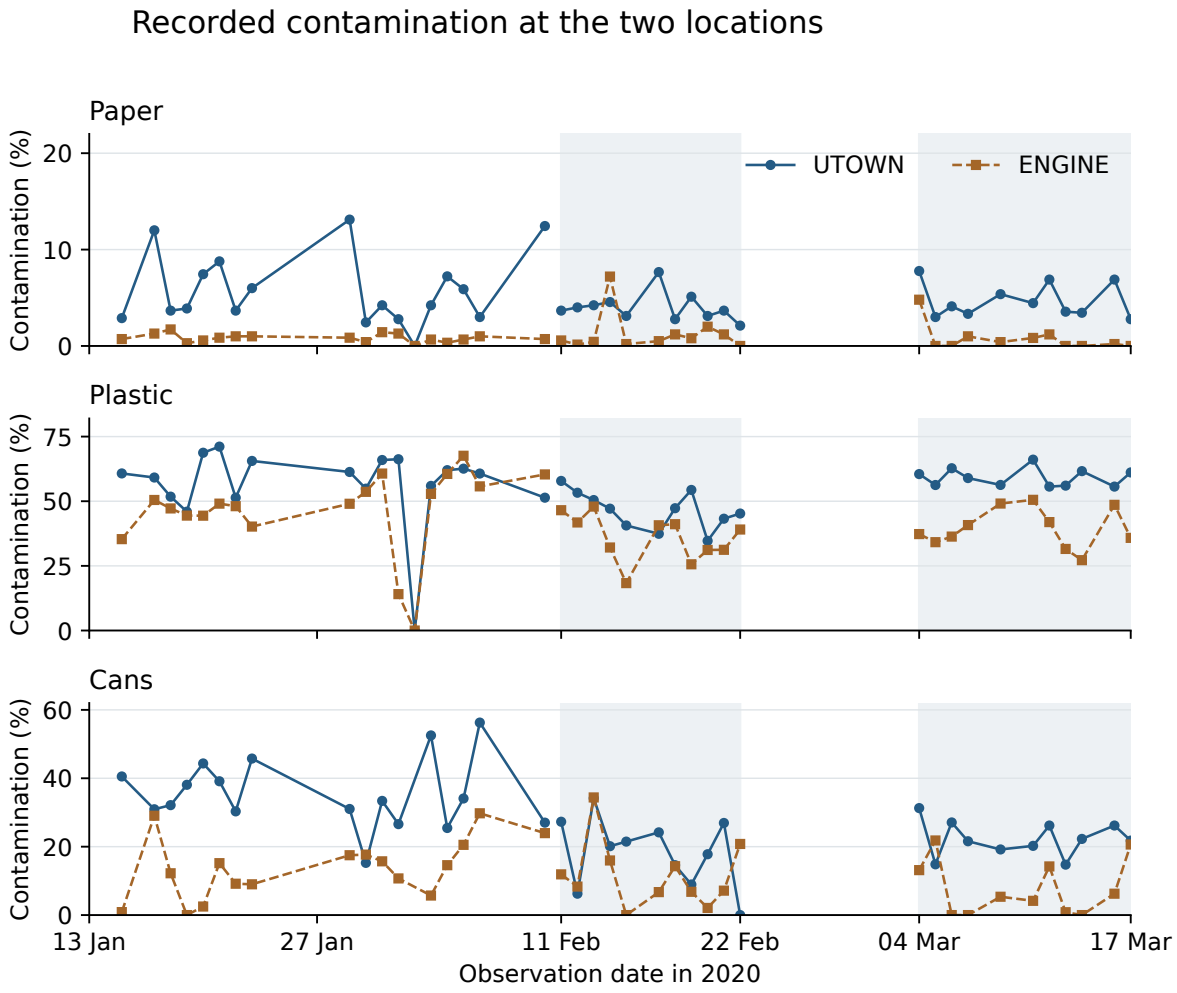


Figure 3: Daily contamination at UTOWN and ENGINE on matched days. Each material has its own vertical scale, and no values are filled in between phases.

C Regression Checks

The intervals in Table 2 come from ordinary regression, which assumes that the errors are independent and equally spread. Figure 4 shows each day's reading minus its area and phase average, on the same vertical scale within each material.

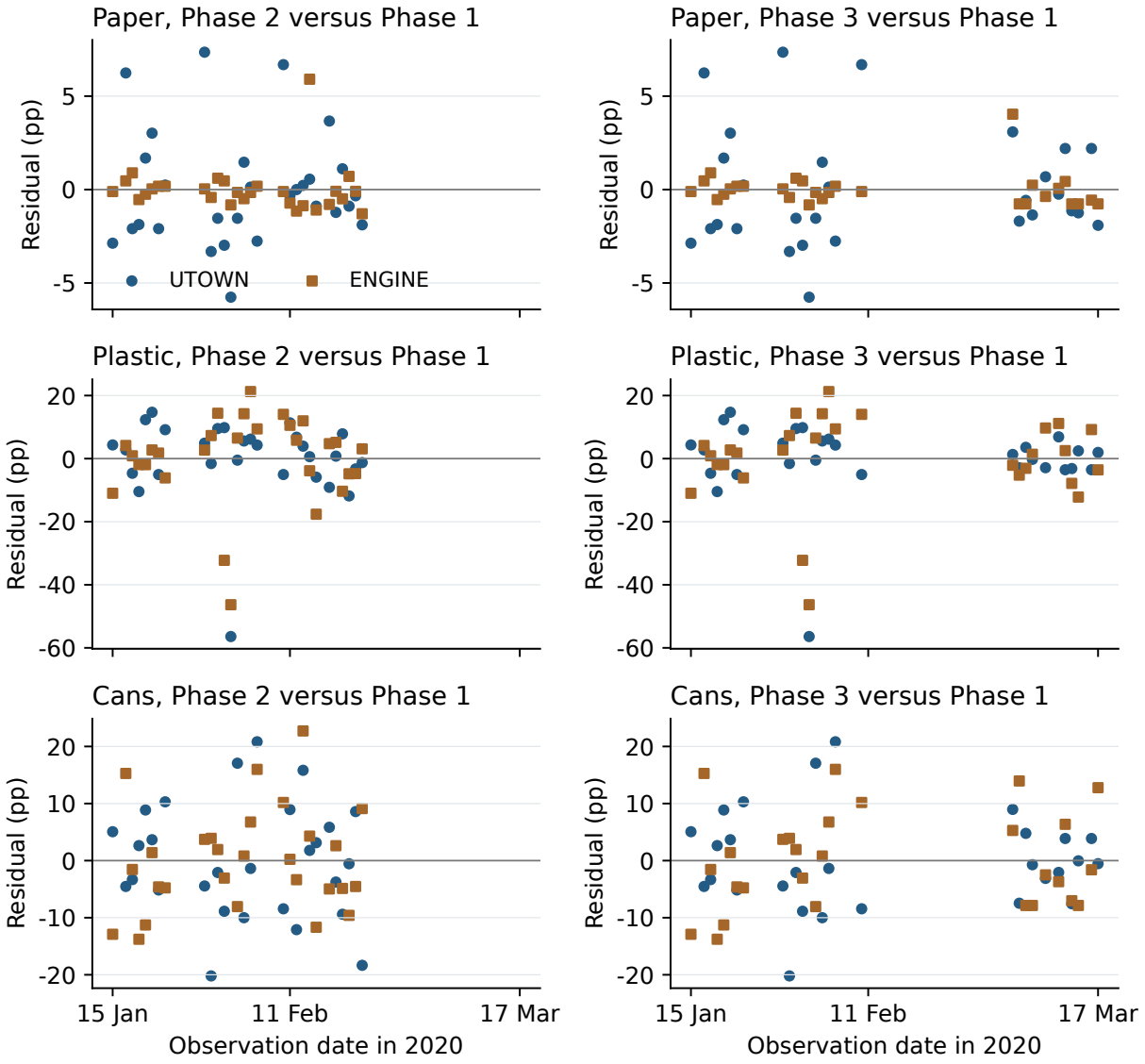


Figure 4: Residuals from the six DiD regressions.

The spread differs between groups, particularly for paper, and readings from the same bins on nearby days may be related. With only two areas, each day cannot be treated as a separate trial. For these reasons the main report gives as much weight to the size of the estimates, the baseline check and the robustness checks as to the p-values.

D Course Links and References

The analysis follows the course sequence. It defines the question, examines the data and the comparison area, estimates the DiD and draws practical implications.

Analysis step	Course connection
Baseline gaps and placebo check	Lecture 5 [4], slides 14–15 (movement before treatment)
Four-average DiD	Lecture 5, slides 7–11 and 33 (comparing the two areas' changes)
DiD regression	Lecture 5, slides 25, 34 and 40 (both main effects and the interaction)
Intervals and residual checks	Basic Statistics [5], slides 48–61 (regression, inference and error assumptions)
Alternative choices of days	Lecture 5, slides 16 and 18–21 (sensitivity and sample coverage)
Suggested further trial	Lecture 4 (random assignment and comparison under shared conditions)

Table 10: How each step of the analysis links to the course.

References

[1] DBA5101. *Group Project 2: Analyzing Recycling Behavior at NUS*. AY2026/2027 assignment brief, pp. 1-2.

[2] DBA5101. *Data Glossary for Project2Data.xls*. Course handout. Data supplied as `GP2Data.xls`.

[3] H. M. Tan, K. A. Too, D. Lin and J. Goh. *Toward Zero Waste: RecycleRight at the National University of Singapore*. Ivey Publishing, case W37339, 2024. First trial and Exhibit 5, pp. 6 and 11.

[4] DBA5101. *Lecture 5: Difference-in-Differences*. Course slides, especially slides 7-16, 18-25, 34 and 40.

[5] DBA5101. *Basic Statistics*, updated handout, slides 33-41 and 48-61; *Lecture 2* regression notebook.

[6] DBA5101. *Lecture 4: Randomized Controlled Trials*. Course slides.

[7] National University of Singapore. *NUS in 2020: COVID-19 Mitigation*, p. 2. Official NUS review. Accessed 7 October 2026.

[8] National University of Singapore. *Academic Calendar AY2019/2020*, p. 1. Official NUS calendar. Calendar text checked 7 October 2026.

[9] S. Duffy and M. Verges. It matters a hole lot: Perceptual affordances of waste containers influence recycling compliance. *Environment and Behavior*, 41(5):741–749, 2009. doi:10.1177/0013916508323737.

[10] J. Austin, D. B. Hatfield, A. C. Grindle and J. S. Bailey. Increasing recycling in office environments: The effects of specific, informative cues. *Journal of Applied Behavior Analysis*, 26(2):247–253, 1993. doi:10.1901/jaba.1993.26-247.

[11] M. Bertrand, E. Duflo and S. Mullainathan. How much should we trust differences-in-differences estimates? *The Quarterly Journal of Economics*, 119(1):249–275, 2004. doi:10.1162/003355304772839588.